

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=other

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	other	silpo	promo_controlled	nan	nan
prozorro	0.9029715543675233	0.01	1.3734510184088137e-14	0.1	1.2601733598025704e-10	0.1	I(1)	1.0	other	silpo	promo_controlled	nan	nan
retail	0.9265122856099318	0.06750725547200247	4.000169254576139e-06	0.1	6.918723434724411e-05	0.041666666666666783	I(1)	0.0	other	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	other	silpo	promo_controlled	nan	0.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	other	silpo	promo_controlled	3.702246334386687e-06	50.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	other	silpo	promo_controlled	nan	0.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
other	silpo	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	50	5.213950941002075	3.4576682875254376	nan	nan	nan	nan	nan	unreliable	1	bounds_lower_p=0.0002606;...	Primary chain is strictly...
other	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	48	0.9816109260310837	-9.345839147206291	-0.29588648930816513	0.025883762971735388	0.5610277733165165	0.26078751490587226	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	prozorro_to_retail	retail	prozorro	0.018450979329619817	0.6777063249902414	0.06806309838224132	1
NARDL	prozorro_to_retail	retail	prozorro	0.07342148057897084	0.2666931649790291	1.3814940607279878e-81	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2025-04-14	nan	5.0	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-15	nan	5.295	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-16	nan	4.985	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-17	nan	5.43	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-18	nan	5.100000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-19	nan	5.175000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-21	nan	5.4	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-22	nan	5.25	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-23	nan	5.0600000000000005	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-24	nan	5.235	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-25	nan	5.46	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-28	nan	5.2	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-29	nan	4.72	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-04-30	nan	4.775	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-01	nan	5.08	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-02	nan	5.05	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-05	nan	5.100000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-06	nan	4.995	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-07	nan	5.22	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-08	nan	5.04	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-09	nan	4.675000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-12	nan	4.78	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-13	nan	4.890000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-14	nan	5.2	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-15	nan	4.95	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-16	nan	5.04	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-19	nan	4.715	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-20	nan	4.545	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-21	nan	4.59	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-22	nan	4.77	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-23	nan	5.265000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-25	nan	4.5	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-26	nan	5.33	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-27	nan	5.55	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-28	nan	4.56	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-29	nan	4.98	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-05-30	nan	4.65	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-06-02	nan	4.66	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-06-03	nan	4.700000000000001	nan	other	silpo	promo_controlled	daily_median_of_available...
2025-06-04	nan	4.7	nan	other	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
prozorro_to_retail	1	0.00916800211923433	
prozorro_to_retail	2	-0.04391836570778855	
prozorro_to_retail	3	-0.09898741784622274	
prozorro_to_retail	4	-0.1737132624652432	
prozorro_to_retail	5	-0.33876966850034373	
prozorro_to_retail	6	-0.33942308793212794	
prozorro_to_retail	7	-0.3492779064865009	
prozorro_to_retail	8	-0.41054888318660143	
prozorro_to_retail	9	-0.3720619865418977	
prozorro_to_retail	10	-0.3444874886934117	
prozorro_to_retail	11	-0.33455886476950003	
prozorro_to_retail	12	-0.28427038150998307	
prozorro_to_retail	13	-0.23827556616559672	
prozorro_to_retail	14	-0.17170479796611818	
prozorro_to_retail	15	-0.20245078372142067	
prozorro_to_retail	16	-0.22691770411699	
prozorro_to_retail	17	-0.18171814006552792	
prozorro_to_retail	18	-0.24953730761858695	
prozorro_to_retail	19	-0.22857510107389625	
prozorro_to_retail	20	-0.2569008753839704	
prozorro_to_retail	21	-0.19322411164274927	
prozorro_to_retail	22	-0.17752826301030095	
prozorro_to_retail	23	-0.1607703008590289	
prozorro_to_retail	24	-0.188453881322739	
prozorro_to_retail	25	-0.24972385400531755	
prozorro_to_retail	26	-0.2925937552753249	
prozorro_to_retail	27	-0.3321217486166532	
prozorro_to_retail	28	-0.32435177197144144	
prozorro_to_retail	29	-0.3501851987733172	
prozorro_to_retail	30	-0.32088012141607464	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
other	silpo	promo_controlled	daily	ambiguous	I(1)	I(1)	0	232	54				Low pair overlap in main ...

Table: NARDL_Multipliers

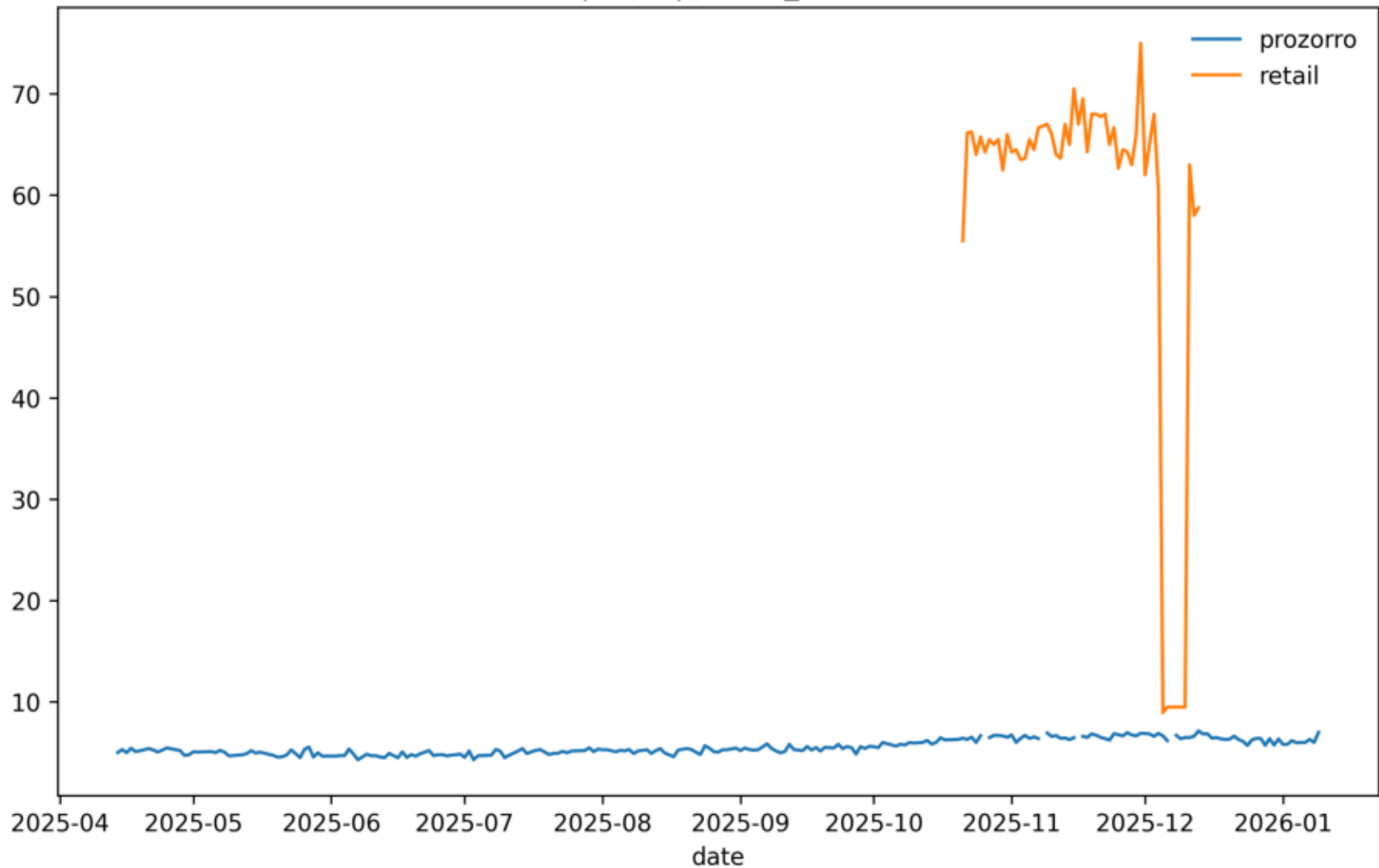
horizon	mult_pos		mult_neg	link
0	-2.656845491478793	-3.6384564175098766	prozorro_to_retail	
1	-2.833764414561026	-3.880740657648056	prozorro_to_retail	
2	-2.84554541780151	-3.896874327062601	prozorro_to_retail	
3	-2.846329912952929	-3.8979486655694284	prozorro_to_retail	
4	-2.84638215236158	-3.898020205601494	prozorro_to_retail	
5	-2.8463856309755795	-3.8980249694409688	prozorro_to_retail	
6	-2.8463858626159384	-3.8980252866642893	prozorro_to_retail	
7	-2.846385878040833	-3.8980253077881404	prozorro_to_retail	
8	-2.8463858790679746	-3.8980253091947743	prozorro_to_retail	
9	-2.846385879136372	-3.8980253092884416	prozorro_to_retail	
10	-2.8463858791409264	-3.8980253092946793	prozorro_to_retail	
11	-2.8463858791412298	-3.8980253092950945	prozorro_to_retail	
12	-2.8463858791412497	-3.898025309295122	prozorro_to_retail	
13	-2.846385879141251	-3.898025309295124	prozorro_to_retail	
14	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	
15	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	
16	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	
17	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	
18	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	
19	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	
20	-2.846385879141251	-3.8980253092951243	prozorro_to_retail	

Table: VECM_IRF

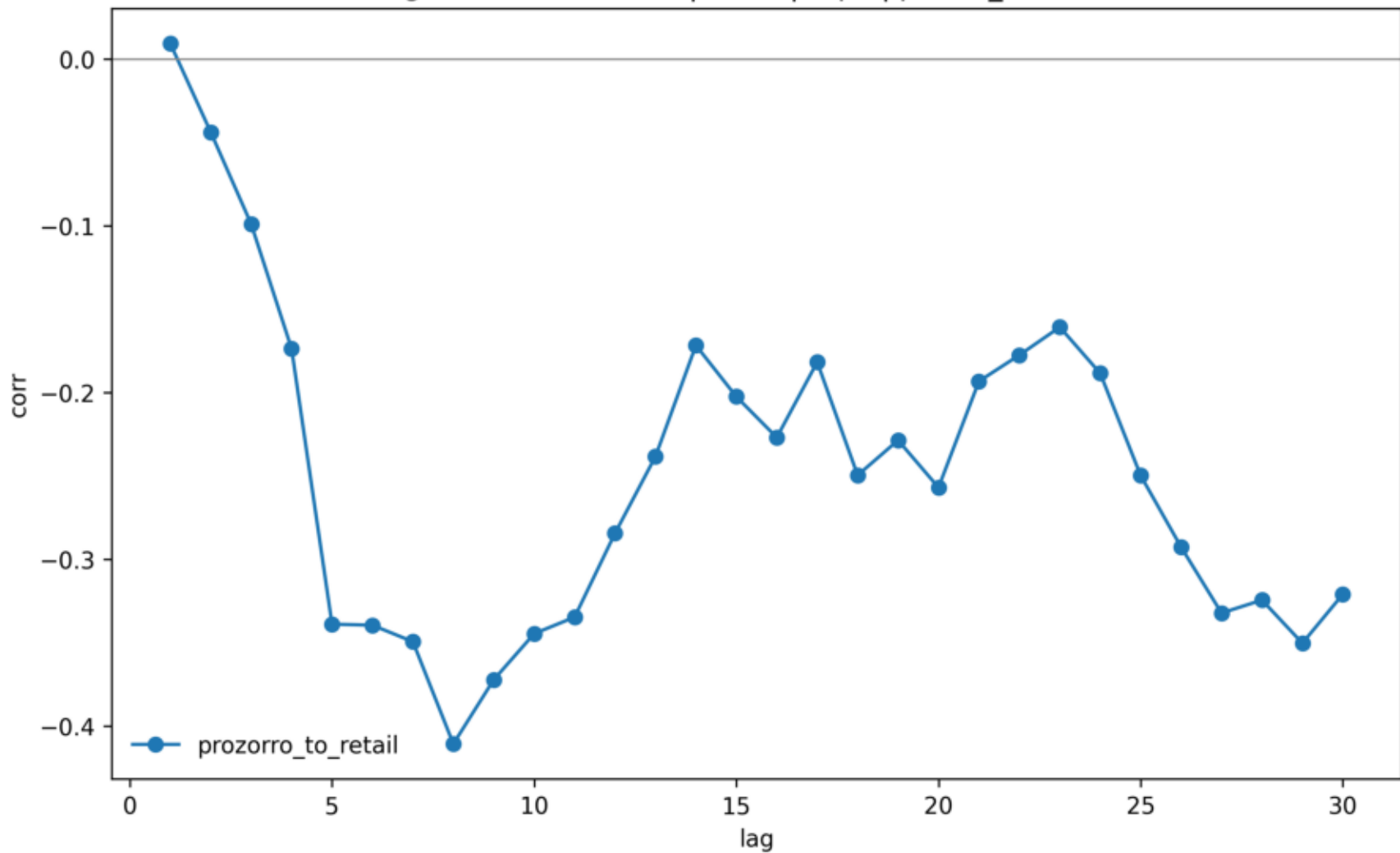
No rows

time_series_promo_controlled.png

other | silpo | promo_controlled



Lag-Correlation Profile | other | silpo | promo_controlled



NARDL Dynamic Multipliers | other | silpo | promo_controlled

